# Analyzing the Relationship between Physical Activity and Mental Health in Adults

By Ryan Shea & Dawson Ullrich

# Paper Analysis

- We looked at three papers:
  - Physical Activity and Mental Health by Scott A. Paluska and Thomas L. Schwenk
  - Mental health benefits of physical activity by S. Saxena, M. Van Ommeren, K. C. Tang and T. P. Armstrong
  - Physical activity and mental health in a student population by Philip Tyson ,Kelly Wilson, Diane Crone ,Richard Brailsford and Keith Laws
- Paluska and Schwenk found that physical activity in adults can be beneficial to adults' mental health, likening the benefits to that of meditation or relaxation. However, they also found that too much exercise can negatively impact mental health..
- Saxena found that physical activity helped with the treatment of common mental health disorders.
- Tyson found that among a student population, those who exercised more exhibited better mental health.
- However, it is important to note that there was not strong evidence of a large effect of exercise on mental health in these findings

# 2023 BRFSS Dataset: About

- We are using the 2023 Behavioral Risk Factor Surveillance System (BRFSS).
- It is the world's largest continuously conducted health survey, created by the CDC.
- Collects data from 400,000+ adults across 78 states and territories.
- Covers a wide range of topics: physical/mental health, healthcare access, chronic conditions, lifestyle habits, and more.

BRFSS™
Behavioral Risk Factor Surveillance System

# 2023 BRFSS Dataset: Data Collection

- Uses a multistage sampling design, selecting participants through several layers of randomization.
- Randomization factors include phone type, demographic characteristics, and geographic region.
- Data is weighted using iterative proportional fitting (raking) to accurately represent each state and territory's population.

# 2023 BRFSS Dataset: Why it's Useful

- Contains thousands of data points.
- Includes hundreds of questions, enabling deep analysis of health behaviors and patterns.
- Offers a large, nationally representative sample, ideal for robust statistical modeling.

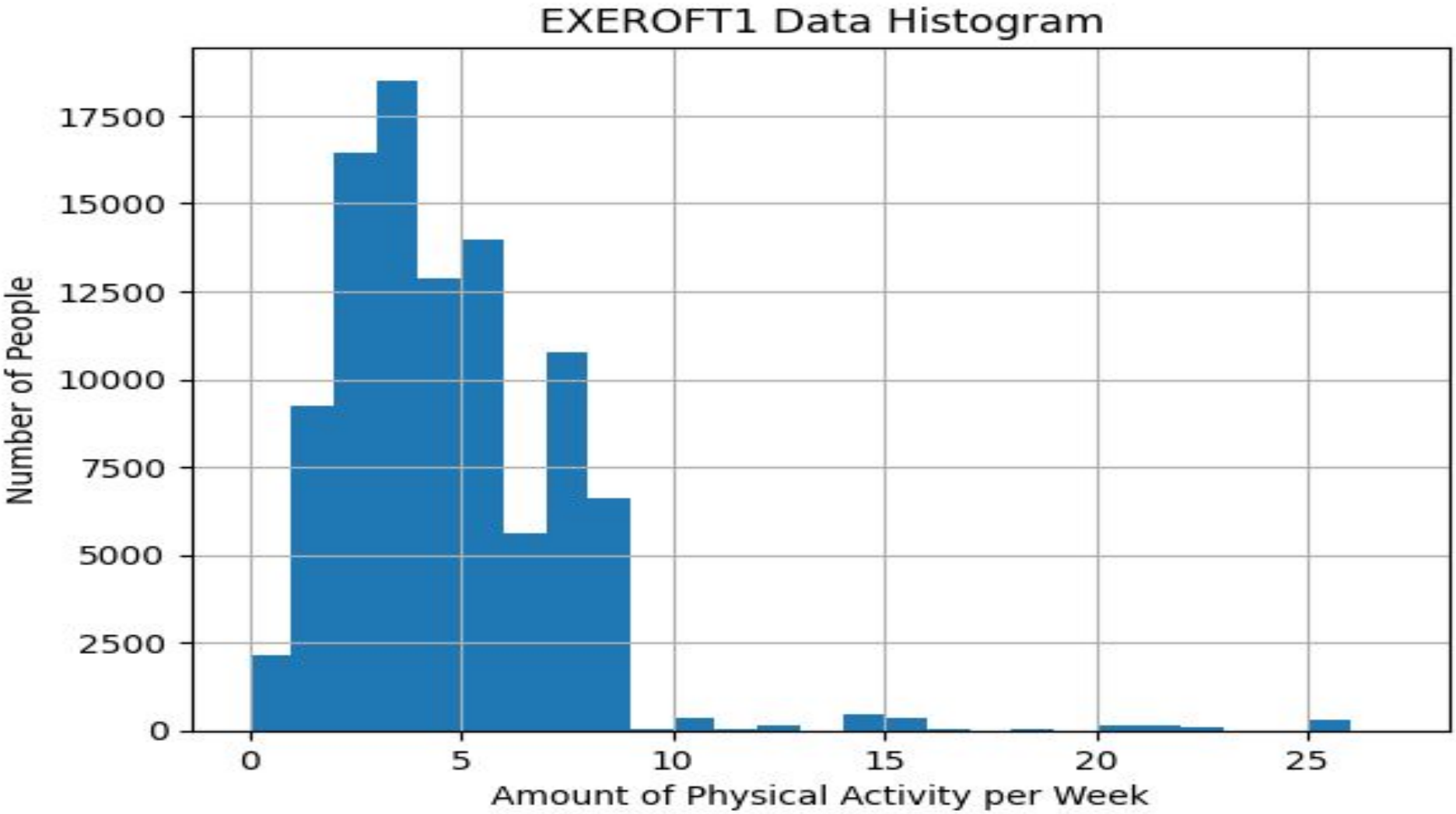
# Initial Dataset Cleaning

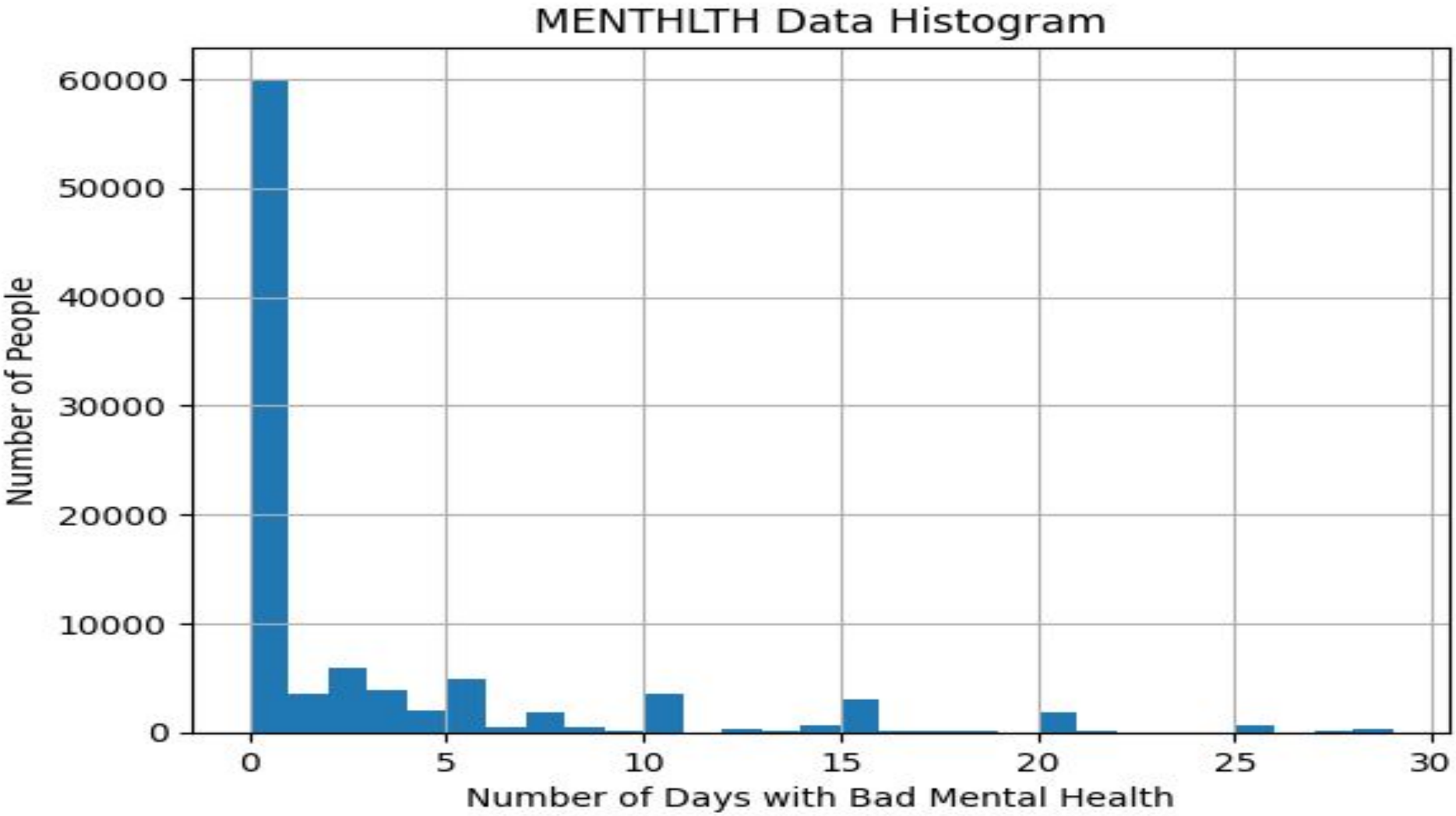
- We looked at two factors: EXEROFT1, and MENTHLTH
  - EXEROFT1: How many times per week or per month did you take part in this activity during the past month?
  - Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?
- First we got nid of all null values in both datasets
- This made is useful for doing more transformation of the dataset

# Dataset Cleaning

- For the EXEROFT1 data, we converted the data into the numbers of times the participants do physical activity per week.  We then eliminated some of the extreme outliers which we determined was over 28 times per week.
- For the MENTHLTH data, we converted the data points with 88 to 0 as it was labeled “None” in the original dataset. We then capped the the maximum amount of days a person could have “not good” mental health was 30.

| | Index | EXEROFT1 | MENTHLTH |
|---|---|---|---|
| count | 57122.000000 | 57122.000000 | 57122.000000 |
| mean | 39650.708466 | 4.270369 | 3.800760 |
| std | 22640.739815 | 2.940308 | 7.557393 |
| min | 1.000000 | 0.000000 | 0.000000 |
| 25% | 19809.250000 | 2.000000 | 0.000000 |
| 50% | 40027.500000 | 4.000000 | 0.000000 |
| 75% | 59138.750000 | 6.000000 | 4.000000 |
| max | 79265.000000 | 28.000000 | 30.000000 |


EXEROFT1 Data Histogram
Number of People
17500
15000
12500
10000
7500
5000
2500
0
0
5
10
15
20
25
Amount of Physical Activity per Week


MENTHLTH Data Histogram
Number of People
60000
50000
40000
30000
20000
10000
0
0
5
10
15
20
25
30
Number of Days with Bad Mental Health

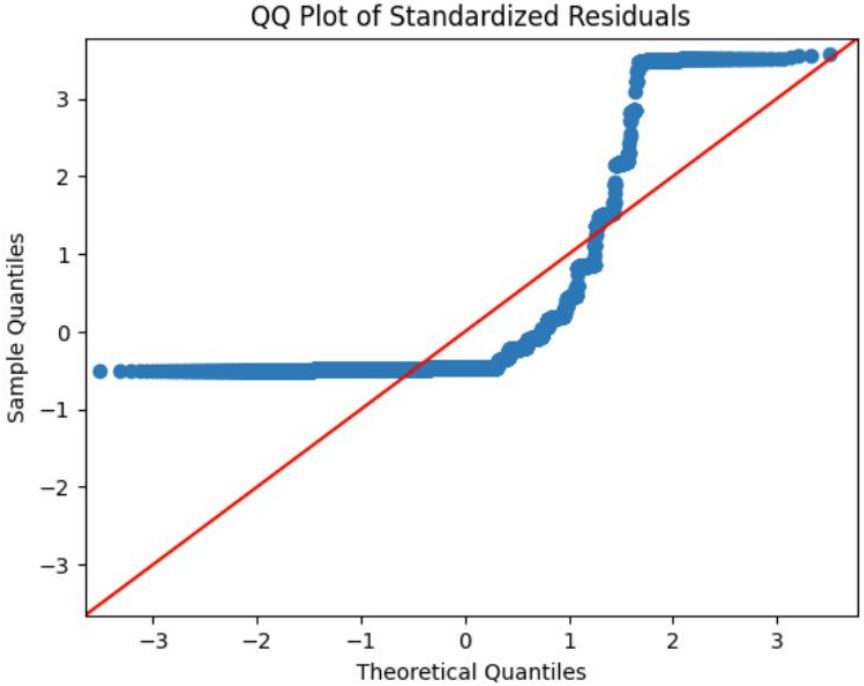
# Regression Analysis

- We ran a simple linear regression model between EXEROFT1 and MENTHLTH and found that the P-Value was 0.061 meaning it is not statistically significant at the 0.05 level.

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **const** | 3.7893 | 0.043 | 87.491 | 0.000 | 3.704 | 3.874 |
| **EXEROFT1** | 0.0158 | 0.008 | 1.872 | 0.061 | -0.001 | 0.032 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 45096.391 | **Durbin-Watson:** | 1.985 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 183073.155 |
| **Skew:** | 2.364 | **Prob(JB):** | 0.00 |
| **Kurtosis:** | 7.728 | **Cond. No.** | 9.34 |

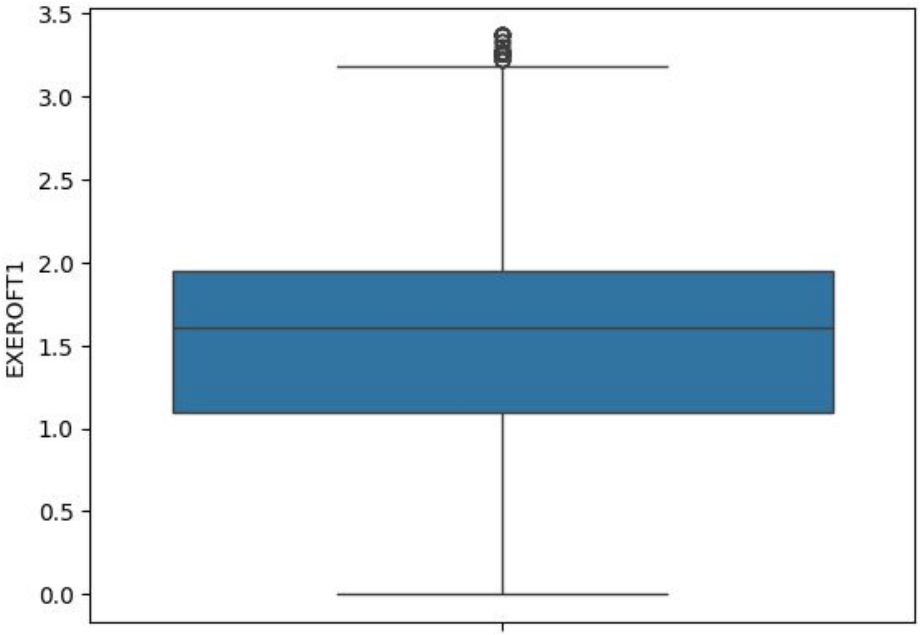
# QQ Pl for OLS Model



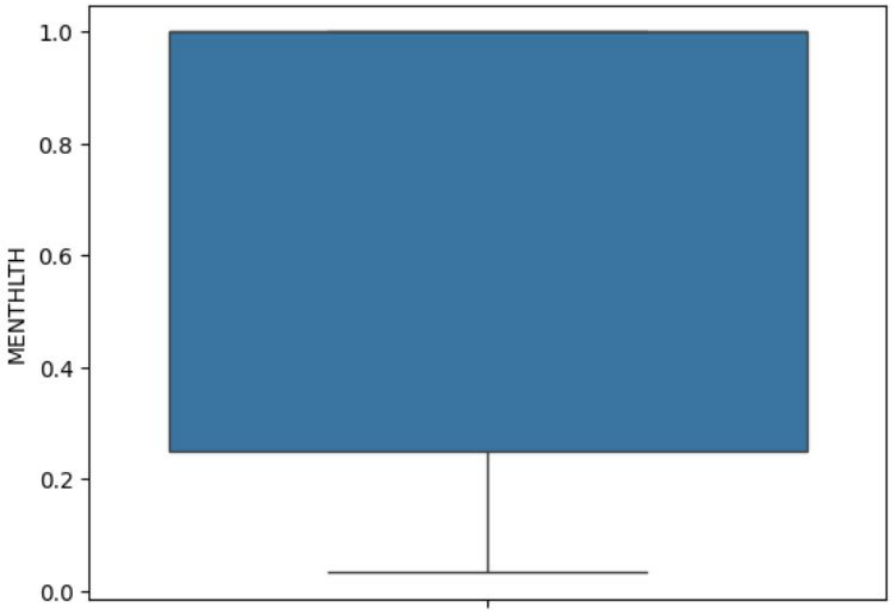
# Variable Transformations

- Because the QQ plot shows that the OLS model violates the normality assumption, we decide to transform the response and independent variable.
- We used the box-cox method to decide on the transformations. This method suggested transforming EXEROFT1 by taking the log of it, and it suggested that we transform MENTHLTH by taking the inverse.

# Boxplot of log(EXEROFT1)



# Boxplot of 1/MENTHLTH



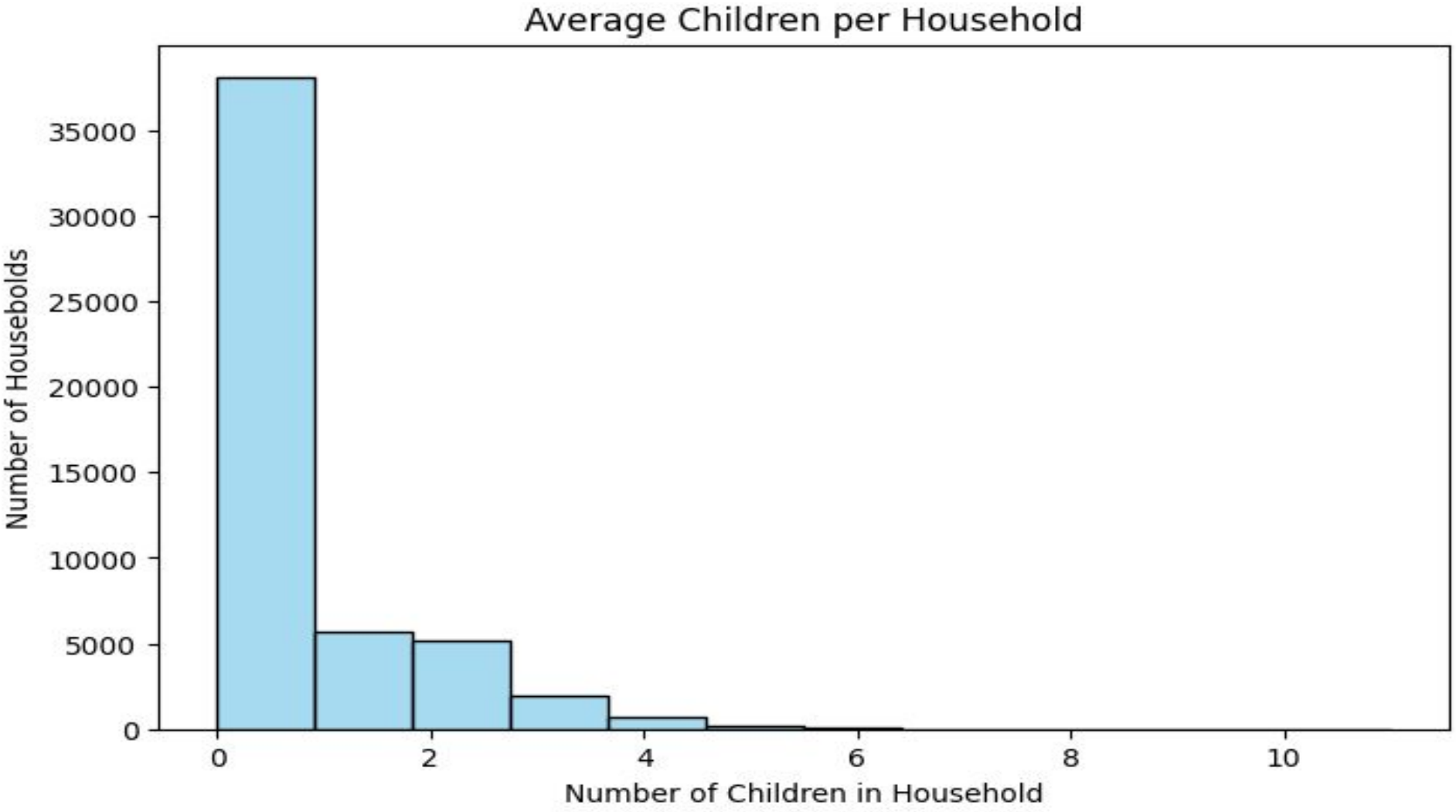
# Control Variable

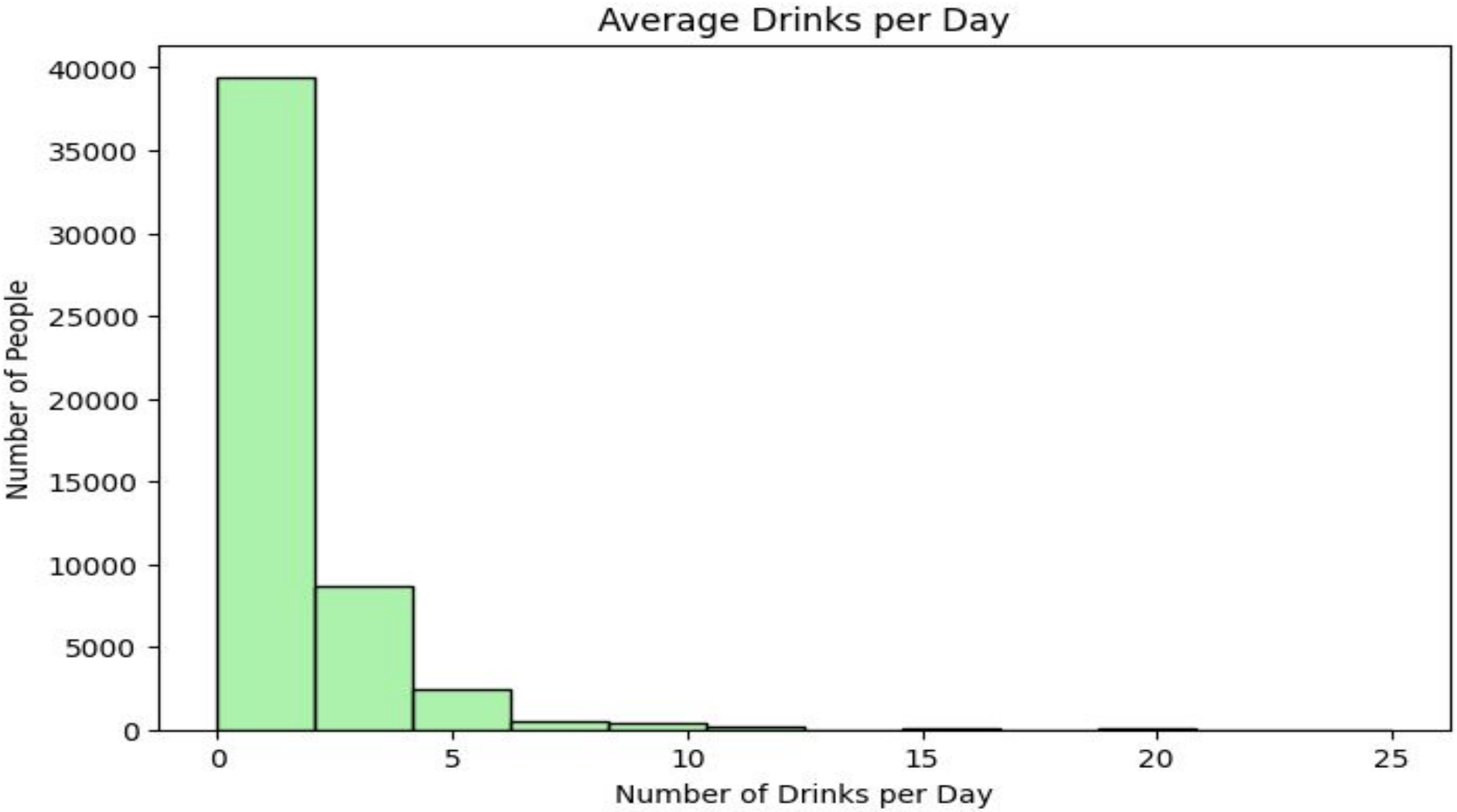
- We then added three control variable to isolate the effect of our main predictor and to avoid misleading results.
- The three control variables are:
  - AVEDRNK3: During the past 30 days, on the days when you drank, about how many drinks did you drink on the average?
  - CHILDREN: How many children less than 18 years of age live in your household?
  - EMPLOY1: Are you currently employed?

# Control Variables Cleaning

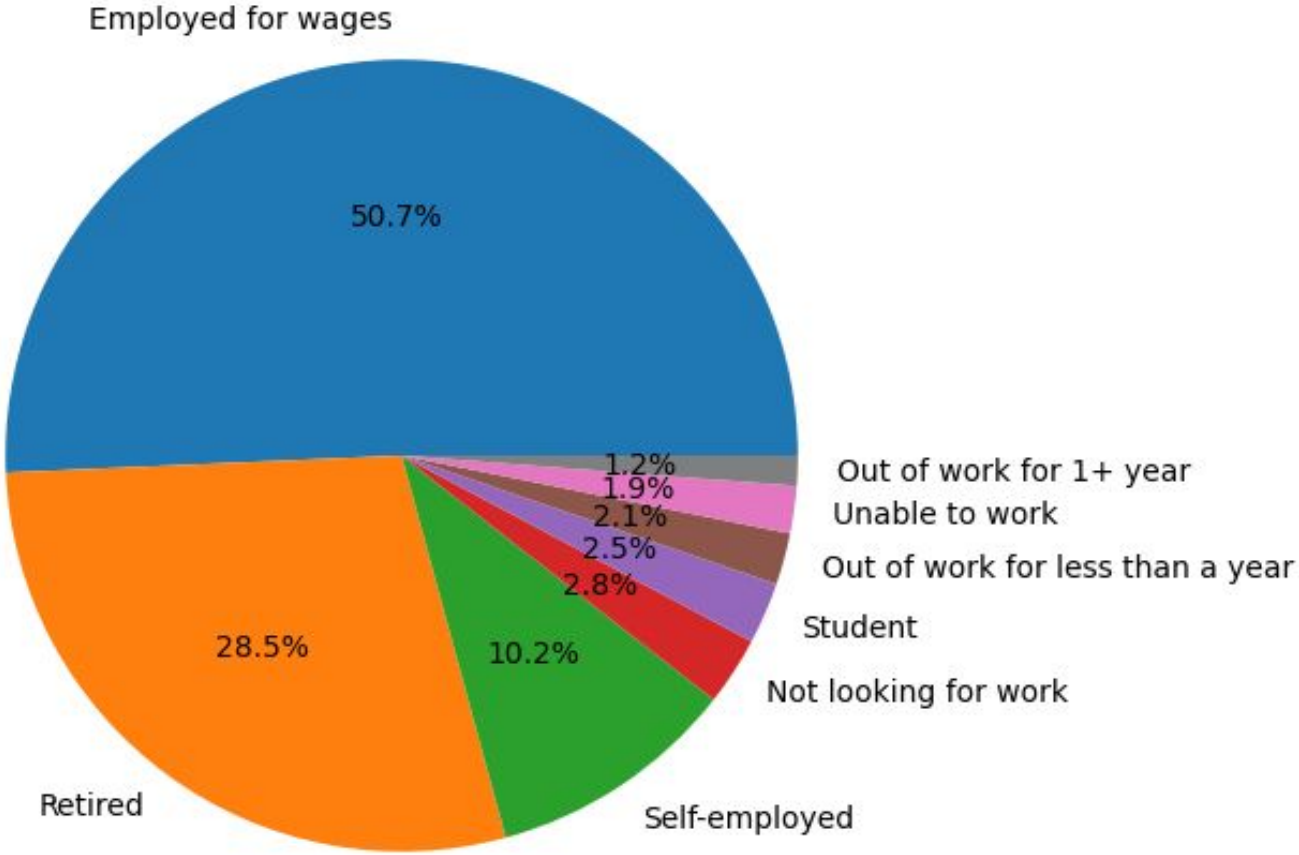
- AVEDRNK3 has data ranging from 1 -78 when people said they did drink. We eliminated the outliers which we determined was over 25 drinks for one day of drinking. We then put the value 88 to 0 as it is labeled “none” in the dataset.
- CHILDREN has data ranging from 1 - 87 of children living at home. We eliminated the outliers which we determined was over 20. We then put the value 88 to 0 as it is labeled “none” in the dataset.
- EMPLOY1 has many different values associated with different forms of employment. We then transformed the data where it only gives back the people who are currently employed, retired or a student.

| | Index | EXEROFT1 | MENTHLTH | bc | AVEDRNK3 | CHILDREN | EMPLOY1 | Unemployed |
|---|---|---|---|---|---|---|---|---|
| **count** | 30551.000000 | 30551.000000 | 30551.000000 | 30551.000000 | 30551.000000 | 30551.000000 | 30551.000000 | 30551.000000 |
| **mean** | 40509.071749 | 4.152466 | 3.799581 | -0.035695 | 2.031128 | 0.442801 | 3.422998 | 0.055612 |
| **std** | 22181.981320 | 2.842199 | 7.327320 | 0.975669 | 1.641976 | 0.914796 | 2.763455 | 0.229175 |
| **min** | 4.000000 | 0.000000 | 0.000000 | -2.715235 | 0.000000 | 0.000000 | 1.000000 | 0.000000 |
| **25%** | 20901.500000 | 2.000000 | 0.000000 | -0.848383 | 1.000000 | 0.000000 | 1.000000 | 0.000000 |
| **50%** | 41996.000000 | 4.000000 | 0.000000 | 0.117688 | 2.000000 | 0.000000 | 2.000000 | 0.000000 |
| **75%** | 59189.500000 | 5.000000 | 4.000000 | 0.478804 | 2.000000 | 0.000000 | 7.000000 | 0.000000 |
| **max** | 79262.000000 | 28.000000 | 30.000000 | 3.995391 | 25.000000 | 10.000000 | 8.000000 | 1.000000 |


Average Children per Household
Number of Housebolds
35000
30000
25000
20000
15000
10000
5000
0
0
2
4
6
8
10
Number of Children in Household


Average Drinks per Day
Number of People
40000
35000
30000
25000
20000
15000
10000
5000
0
0
5
10
15
20
25
Number of Drinks per Day

# Employee Classification Pie Chart



| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **const** | 0.6093 | 0.041 | 14.974 | 0.000 | 0.530 | 0.689 |
| **EXEROFT1** | 0.0380 | 0.016 | 2.323 | 0.020 | 0.006 | 0.070 |
| **AVEDRNK3** | -0.0765 | 0.021 | -3.568 | 0.000 | -0.119 | -0.034 |
| **CHILDREN** | -0.0886 | 0.019 | -4.638 | 0.000 | -0.126 | -0.051 |
| **EMPLOY1** | 0.0952 | 0.014 | 6.713 | 0.000 | 0.067 | 0.123 |
| **Unemployed** | -0.3755 | 0.054 | -6.901 | 0.000 | -0.482 | -0.269 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 21022.128 | **Durbin-Watson:** | 1.983 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 247.411 |
| **Skew:** | -0.431 | **Prob(JB):** | 1.88e-54 |
| **Kurtosis:** | 1.586 | **Cond. No.** | 16.5 |

# Multiple Regression with Controls

- The model is statistically valid but explains only ~7% of the variation in the data, $R^2 = 0.07$.
- Log(EXEROFT1) has a p-value of 0.02 meaning it is statistically significant at the 0.05 level.
- The coefficient of log(EXEROFT1) is positive. This means that as EXEROFT1 increases, 1/MENTHLTH increases. So, MENTHLTH decreases. Thus, more exercise causes predicted mental health to improve.
- All control variables are also statistically significant.

# Instrumental Variable Model

- IV models use an exogenous instrument, a variable that is correlated with the independent variable and isn't correlated with the response variable. IV models allow us to check for possible reverse causality.
- We used height as our exogenous instrument since taller people tend, on average, to have, higher athletic participation, more comfort in physical activities and sometimes better physical capability.
- Height generally does not directly cause mental health changes in adults. So the exclusion restriction can be argued depending on context.
- For cleaning we changes all the entries into inches and eliminated the outliers of over 86 inches and under 53 inches.

- Height is used as an instrument to isolate the causal part of exercise not driven by hidden factors like motivation or baseline health.
- The p-value of log(EXEROFT1) is 0.375, this means it is not statistically significant. Therefore, we can't rule out the possibility of reverse causality.
- The very low f-statistic for the model suggests that height is not a strong exogenous instrument.

```
                        IV-2SLS Estimation Summary
==============================================================================
Dep. Variable:           MENTHLTH_inv   R-squared:                     -22.504
Estimator:                    IV-2SLS   Adj. R-squared:                -22.510
No. Observations:                4372   F-statistic:                    0.7871
Date:                Thu, Nov 27 2025   P-value (F-stat)                0.3750
Time:                        03:39:24   Distribution:                  chi2(1)
Cov. Estimator:                robust

                             Parameter Estimates
================================================================================
              Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
--------------------------------------------------------------------------------
Intercept        6.0299     6.0099     1.0033     0.3157     -5.7492      17.809
EXEROFT1_log    -3.5223     3.9702    -0.8872     0.3750     -11.304      4.2592
================================================================================
```

# Conclusion

- Across the regression models, there is evidence that exercise does have a positive effect in mental health. However, that evidence is relatively weak and only suggest that there is a marginal effect.
- The IV model was unable to rule out the possibility of reverse causality between EXEROFT1 and MENTHLTH.
- Our findings were consistent with research done in the past that showed exercise only has a marginal beneficial effect on mental health.

# References

Paluska, S.A., Schwenk, T.L. Physical Activity and Mental Health. Sports Med 29, 167–180 (2000). https://doi.org/10.2165/00007256-200029030-00003

Saxena, S., Van Ommeren, M., Tang, K. C., & Armstrong, T. P. (2005). Mental health benefits of physical activity. Journal of Mental Health, 14(5), 445–451. https://doi.org/10.1080/09638230500270776

Tyson, P., Wilson, K., Crone, D., Brailsford, R., & Laws, K. (2010). Physical activity and mental health in a student population. Journal of Mental Health, 19(6), 492–499. https://doi.org/10.3109/09638230902968308